

Quick Check

Lesson 7: Dichotomous keys for animals and viruses

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.3: Taxonomy II | Lesson 7

A visitor from Kisumu has never seen the animals at your school. You can only send a written message so that the visitor names the same animal as you. Which statement would work best in the message?

A It is a beautiful animal.

B It is a big animal.

C It has six jointed legs.

D It is a useful animal for food.

QUESTION 2 OF 8

A group builds this part of a dichotomous key.

1a. Animal has a backbone: go to step 2.

1b. Animal has no backbone: go to step 3.

2a. Feathers present: chicken.

2b. Feathers absent, hair present: goat.

3a. Six jointed legs present: grasshopper.

3b. No legs, body covered by a hard shell: snail.

A

A Goat

B Grasshopper

C Chicken

D Snail

A group wrote this couplet: 'Animal is small' / 'Animal is big'. After swapping keys, the testing group got different answers from the authors. What is the main problem with this couplet?

- A** Animals never differ in size, so the feature is useless.
- B** The couplet should name the animal in the first statement.
- C** A couplet must have three choices, not two.
- D** Two people can disagree about what small and big mean, because the statements are not measurable or countable.

Learners sort animals into Phylum Chordata or Phylum Arthropoda. Which list contains only Arthropoda?

- A** Chicken, lizard, grasshopper
- B** Butterfly, earthworm, grasshopper
- C** Mosquito, butterfly, grasshopper
- D** Mosquito, butterfly, snail

Which statement correctly describes a virus?

- A** It has a nucleus and mitochondria, like the cells of a goat.
- B** It is a very small bacterium that divides on its own in soil.
- C** It is a non-cellular particle with a protein coat around DNA or RNA, and it reproduces only inside a host cell.
- D** It carries out respiration and growth by itself outside a host.

A learner claims that a virus is not a living organism. Which piece of evidence best supports this claim?

- A** A virus has genetic material, DNA or RNA.
- B** A virus can cause disease in maize and in people.
- C** A virus is very small.
- D** A virus has no cell structure and cannot reproduce or carry out metabolism outside a host cell.

A health worker says that antibiotics will not cure measles. What is the correct reason?

- A** Measles is caused by a very large bacterium that resists antibiotics.
- B** Antibiotics only work on diseases of plants.
- C** Measles is not caused by any organism or particle.
- D** Measles is caused by a virus, which is not a bacterium, and antibiotics work against bacteria.

Learners add a box for viruses to the class model. Which example shows a useful (beneficial) importance of viruses?

- A** Bacteriophages being used to attack harmful bacteria
- B** Cassava mosaic disease reducing cassava yields on Kenyan farms
- C** Foot and mouth disease in cattle
- D** Newcastle disease killing chickens

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).